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Research paper

Alterations in gray matter volume and cortical thickness in medication-naive offspring with ADHD and a parental history of bipolar disorder

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ABSTRACT

Background: Bipolar disorder (BD) is a high heritability rate mental disorder. Attention-deficit/hyperactivity disorder (ADHD) in offspring of individuals with BD may manifest as a prodromal stage of BD and have distinct neuroimaging features compared to typical ADHD.
Methods: We conducted a cross-sectional involving 101 medication-naive participants aged 8 to 15 years dividing into three groups: BD offspring with ADHD ($n = 21$), non-BD offspring with ADHD ($n = 36$), and healthy controls ($n = 44$). Voxel-based morphometry and Surface-based morphometry methods were used to calculated gray matter volume and cortical thickness of three groups. After conducting the aforementioned analyses, we pooled all ADHD patients into a single cohort and conducted correlation analyses between the SNAP-IV subscale scores and the mean values of differential brain regions.
Results: Significant reductions in gray matter volume were observed in BD offspring within the limbic system compared to both other groups. Cortical thickness thinning was found in bilateral precentral gyri, bilateral fusiform gyri, bilateral parahippocampal gyri, bilateral temporal pole in BD offspring compared to both other groups. Non-BD offspring showed decreased cortical thickness in the bilateral precentral gyrus when compared to HCs. The severity of ADHD hyperactivity/impulsivity symptoms is negatively correlated with cortical thickness in the precentral gyrus.
Conclusions: Offspring of individuals with BD diagnosed with ADHD exhibit more pronounced brain structural alterations than those with typical ADHD, suggesting these changes may represent the genetic abnormalities underlying bipolar disorder.

1. Introduction

Bipolar disorder (BD) is a chronic mood disorder characterized by recurrent episodes of depression and mania/hypomania with a worldwide prevalence of approximately 1 % (Merikangas et al., 2011). BD is one of the top 20 causes of global burden of diseases, which causes great morbidity and disability (Vigo et al., 2016). BD imposes a significant global health burden in 2019, contributing 6.2 million disability-adjusted life-years (DALYs), primarily due to years lived with disability (YLDs), with an age-standardized DALY rate of 14.9 per 100,000 population (GBD 2019 Adolescent Mortality Collaborators, 2021). BD is a type of mental disorder with a high genetic predisposition, with a heritability rate ranging from 60 % to 85 % (Song et al., 2015; Mullins et al., 2021; Bienvenu et al., 2011). Having a first-degree relative with BD significantly elevates the likelihood of developing the condition (Doucette et al., 2016). Children born to parents with BD exhibit heightened susceptibility to developing psychiatric disorders, particularly BD itself (Türközer et al., 2025).

The developing from a genetic high-risk status to the full-blown BD in offspring of individuals with BD represents a developmental

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continuum (Wiltink et al., 2015). The later phase of this prodromal period is often referred to as the ultra-high-risk (UHR) stage, characterized by the presence of subthreshold symptoms that precede a definitive BD diagnosis. Commonly reported prodromal symptoms include irritability, sleep disturbances, hyperactivity, anxiety, and subthreshold manic or depressive episodes (Axelson et al., 2011; Howes and Falkenberg, 2011). There is a significant association between attention-deficit/hyperactivity disorder (ADHD) and the subsequent development of BD in offspring of individuals with BD (Duffy et al., 2014). Notably, childhood ADHD is more prevalent among individuals at high familial risk for BD and has been proposed as a potential early marker or even a prodromal feature of BD in some cases (Raouna et al., 2018). This overlap contributes to the frequent misdiagnosis of BD in its early stages, particularly when symptoms such as inattention — commonly associated with ADHD — dominate the clinical presentation (Lublóy et al., 2020; Vieta et al., 2018). Such diagnostic ambiguity often leads to delays in appropriate intervention, underscoring the importance of identifying specific neurobiological markers that can differentiate between these conditions and aid in early detection among genetically vulnerable populations.

Neuroimaging has increasingly been recognized as a promising source of objective biomarkers in psychiatry. Morphometric studies have identified widespread gray matter (GM) alterations in individuals with BD. For instance, reductions in gray matter volume (GMV) have been consistently reported in the right prefrontal cortex, right temporal cortex, right insula, and limbic structures among BD patients (Long et al., 2023; Selvaraj et al., 2012). Additionally, cortical thickness (CT) analyses have revealed significant thinning in BD compared to healthy controls, particularly in frontoinsular regions such as the left insula, the orbital part of the left inferior frontal gyrus, the medial portion of the left superior frontal gyrus, and the bilateral anterior cingulate cortex (Zhu et al., 2022).In contrast, individuals with ADHD also show reduced GM volume relative to healthy controls, with the most consistent findings observed in subcortical structures, particularly the basal ganglia (Albajara Sáenz et al., 2019; Samea et al., 2019). Cortical thickness abnormalities, most commonly cortical thinning, have been reported across various brain regions, including the frontal, prefrontal, parietal, and temporal cortices; however, these findings tend to be less consistent than those observed in BD (Albajara Sáenz et al., 2019; Samea et al., 2019; You et al., 2024). Importantly, accumulating evidence suggests that structural brain alterations in BD may reflect a progressive neurodevelopmental process. Studies indicate a gradual reduction in gray matter volume within key brain regions such as the limbic system and prefrontal cortex, showing a continuum of decreasing GMV from healthy individuals to those at high genetic risk, and further to those at ultra-high risk for BD (Lin et al., 2015). These findings support the hypothesis that neuroanatomical changes may serve as early indicators of disease progression and could inform early intervention strategies. Notably, studying medication-naive offspring of individuals with BD in the prodromal phase offers important advantages, as it minimizes potential confounding effects of psychotropic medications and comorbid psychiatric conditions on the observed neuroimaging outcomes (Goff et al., 2017; Lai, 2021).

Based on the above studies, we hypothesize that ADHD in offspring of individuals with BD patients may exhibit distinct neuropathological features compared to typical ADHD. The gray matter volume and cortical thickness abnormalities in medication-naive offspring of individuals with BD with ADHD show pattern differences when compared to those observed in individuals with typical ADHD. Specifically, compared to typical ADHD, offspring of individuals with BD with ADHD may exhibit more pronounced structural abnormalities in the brain structures.

2. Methods

2.1. Participants

This study employed a cross-sectional design. Participants included children diagnosed with ADHD who were recruited from the Jiangmen Maternity and Child Health Care Hospital, as well as healthy controls recruited from the local community in Jiangmen City. Three groups of participants were recruited: (1) BD offspring with ADHD group (BDO): Offspring of individuals with BD who were themselves diagnosed with ADHD; (2) Non-BD offspring with ADHD group (NBDO): Children diagnosed with ADHD who had no family history of BD, schizophrenia, schizoaffective disorder, or major depressive disorder; (3) Healthy control group (HC): Typically developing children without any DSM-5 diagnosis. The inclusion criteria for BD offspring were as follows: 1. Diagnosis of ADHD based on DSM-5 criteria; 2. At least one biological parent diagnosed with BD; 3. No prior exposure to neuroleptics, mood stabilizers, or psychostimulant medications. The inclusion criteria for Non-BD offspring were as follows: 1. Diagnosis of ADHD based on DSM-5 criteria; 2. No family history of BD, schizophrenia, schizoaffective disorder, or major depressive disorder; 3. No prior exposure to neuroleptics, mood stabilizers, or psychostimulant medications. The inclusion criteria for healthy controls were: 1. No current or history DSM-5 diagnosis; 2. No family history of mental illness. All participants were aged between 8 and 15 years and were free from thyroid dysfunction, and neurological conditions such as cranioencephalic injuries or other central nervous system diseases. Comorbid diagnoses including autism spectrum disorder (ASD), schizophrenia, bipolar disorder, major depressive disorder, anxiety disorders, obsessive-compulsive disorder (OCD), and substance use disorders were excluded. The diagnosis of ADHD was based on the Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children-Present and Lifetime version 5 (K-SADS-PL-5). All participants underwent clinical assessments using the following scales: Hamilton Depression/Anxiety Rating Scale (HAM-D/A), Young Mania Rating Scale (YMRS), 18-Item Swanson, Nolan and Pelham-IV Rating Scale (SNAP-IV).

This study was approved by the Institutional Review Board (IRB) of Jiangmen Maternity and Child Health Care Hospital. Written informed consent was obtained from the legal guardians of all participants.

2.2. MRI data acquisition

MRI data were collected using a 1.5 T Philips MRI scanner with a standard head coil. Structural brain image data of T1-weighted images were obtained using a magnetization-prepared rapid gradient echo (MPRAGE) sequence. The sequence parameters were as follows: TR = 1035 ms, TE = 5.2 ms, flip angle = 15°, field of view = 256 × 256, voxel size = 1 × 1 × 1 mm^3. Participants were instructed to remain still with their eyes closed during the scan. After the MRI examination completion, radiologists reviewed the image quality and excluded images with significant head motion.

2.3. MRI data processing

Gray matter volume and Cortical thickness were calculated using Voxel-based morphometry (VBM) and Surface-based morphometry (SBM) methods, respectively. Statistical Parametric Mapping 12 package (SPM12, https://www.fil.ion.ucl.ac.uk/spm/) and Computational Anatomy Toolbox 12 (CAT12, http://dbm.neuro.uni-jena.de/cat) both integrated with MATLAB Software were used for VBM and SBM data analysis. Default templates provided by those tools were utilized to normalize the images into Montreal Neurological (MNI) space. After normalization, structural image data were segmented into gray matter (GM), white matter and cerebrospinal fluid.

For VBM analyses, GM data were smoothed using an 8 mm full-width-at-half-maximum (FWHM) Gaussian Kernel. For SBM analyses,

brain surfaces were reconstructed using the projection-based thickness method, an automated surface-preprocessing algorithms implemented in CAT12 (Gaser et al., 2024). Cortical thickness was calculated as the distance between the inner and outer surfaces at each point on the mesh. Prior to further analyses the cortical thickness data were smoothed with a 15 mm FWHM Gaussian Kernel.

2.4. Statistical analysis

Statistical analyses of demographic and clinical data were conducted using R version 4.1.3 (https://cloud.r-project.org/). Parametric (one-way analysis of covariance, ANCOVA) and/or non-parametric (Kruskal-Wallis test or Chi-square test) tests were applied to assess group differences after evaluating the normality of continuous variables. Scheffé method was used for post-hoc pairwise comparisons with a significance level at $p < 0.05$ to adjust for multiple comparisons.

For VBM and SBM data, statistical analyses were performed using SPM12 and CAT12. One-way analysis of covariance (ANCOVA) was employed to examine cross-group differences, with age, sex, and total intracranial volume (TIV) included as covariates. To control the family-wise error rate (FWER), threshold-free cluster enhancement (TFCE) was applied with 5000 random permutations, and statistical significance was set at $p < 0.05$ for family wise error correction (FWE) to correct for multiple comparisons. The anatomical locations of significant clusters showing gray matter volume (GMV) and cortical thickness (CT) differences were determined with reference to the AAL and Desikan-Killiany atlases respectively. After ANCOVA analysis, significant cluster values of individuals were extracted. Group differences were further analyzed using R software, with post-hoc pairwise comparisons conducted via the Scheffé method with a significance level at $p < 0.05$ to control for multiple comparisons and identify specific differences between groups.

Correlation analyses were conducted using R (version 4.1.3). All individuals with ADHD were pooled into a single group, and brain regions showing significant differences in prior group comparisons were defined as regions of interest (ROIs). Within these ROIs, we examined associations between cluster-level values and clinical symptom severity as measured by the SNAP-IV subscales, including inattention and hyperactivity/impulsivity. To control for Type I error due to multiple comparisons, the Bonferroni correction was applied, with the significance threshold set at $p < 0.05$ divided by the total number of correlation tests performed.

3. Result

3.1. Demographics and clinical measures

A total of 101 participants were included in the study, comprising 21 BD offspring with ADHD, 36 individuals with ADHD and no family history of mental disorders, and 44 healthy controls (HC). The demographic and clinical characteristics are presented in Table 1. No significant differences were found in Hamilton Depression Scale (HAMD) or Young Mania Rating Scale (YMRS) scores among the three groups ($p > 0.05$). Both the BD offspring group and the non-BD offspring group showed significantly higher SNAP-IV inattention subscores and hyperactivity/impulsivity subscores compared to the HC group ($p < 0.05$, Scheffé corrected). Additionally, the BD offspring group exhibited higher hyperactivity/impulsivity subscores than the non-BD offspring group ($p < 0.05$, Scheffé corrected) (Table 1).

3.2. Gray matter volume difference

Results of the one-way ANCOVA (corrected for age, sex, and total intracranial volume) revealed significant gray matter volume differences among the three groups after correction ($p < 0.05$, TFCE-FWE corrected). As shown in Fig. 1, these differences were observed in brain regions in the bilateral limbic system (right parahippocampal & amygdala and left amygdala) (Table 2).

Table 1
Difference among groups in demographic and clinical measures.

	BD offspring ($n = 21$)	Non-BD offspring ($n = 36$)	HC ($n = 44$)	F/χ2	P	Post-hoc
Age (years)	13.40 (2.12)	12.87 (2.35)	13.91 (2.54)	1.86	0.16	NA
Sex (M/F)	14:7	26:10	16:7	0.198	0.91	NA
HAMD	0.71 (1.01)	0.61 (0.93)	0.30 (0.80)	2.04	0.13	NA
YAMS	0.86 (1.28)	0.47 (1.16)	0.25 (0.84)	2.346	0.10	NA
SNAP-IV Inattention total score	19.62 (4.41)	19.44 (3.24)	1.70 (1.37)	460.84	0.000	A > C, B > C
Hyperactivity/ Impulsivity total score	18.62 (4.59)	15.92 (5.02)	2.34 (2.03)	177.58	0.000	A > B, A > C, B > C

Post-hoc tests revealed that the BD offspring group exhibited greater lower gray matter volumes compared to both the non-BD offspring group and healthy controls ($p < 0.05$, Scheffé corrected). No significant differences in gray matter volume were found between the non-BD offspring group and the HC group (Fig. 1) (Supplementary Table 1).

3.3. Cortical thickness analysis

For cortical thickness, our results revealed significant differences in the bilateral precentral gyri, bilateral fusiform gyri, bilateral parahippocampal gyri, bilateral temporal pole after correction ($p < 0.05$, TFCE-FWE corrected) (Table 3) (Fig. 2).

Post-hoc analyses demonstrated that the BD offspring group exhibited significantly reduced cortical thickness in these regions compared to both the non-BD offspring group and healthy controls ($p < 0.05$, Scheffé corrected). Furthermore, the non-BD offspring group also showed decreased cortical thickness in the bilateral precentral gyri when compared to HCs ($p < 0.05$, Scheffé corrected). However, no significant differences in cortical thickness were found between the non-BD offspring group and HCs in the bilateral fusiform gyri, bilateral parahippocampal gyri, or left temporal pole (Fig. 3) (Supplementary Table 2).

3.4. Correlations between ROI region and SNAP-IV subscales score

A total of 20 correlations were conducted between the ROI regions and the SNAP-IV subscales and to account for multiple comparisons, we applied the Bonferroni correction, setting the adjusted significance threshold at 0.0025 (0.05/20) for the 20 correlations. As show in the Supplementary Table 3, our study revealed a significant negative correlation between right precentral gyrus cortical thickness and Hyperactivity/Impulsivity scores ($p = 0.001$), while the left precentral gyrus exhibited a trend toward negative association with the same subscale, though this did not survive multiple comparisons correction ($p = 0.013$) (Supplementary Table 3, Supplementary Fig. 1).

4. Discussion

This study investigated medication-naïve BD offspring with ADHD using structural neuroimaging and compared them to individuals with ADHD but without familial risk for BD. Our results indicate significant group differences in both gray matter (GM) volume and cortical thickness. Specifically, patients with ADHD who have a genetic predisposition to BD exhibited more pronounced reductions in GM volume and cortical thickness, even in the absence of overt BD symptoms.

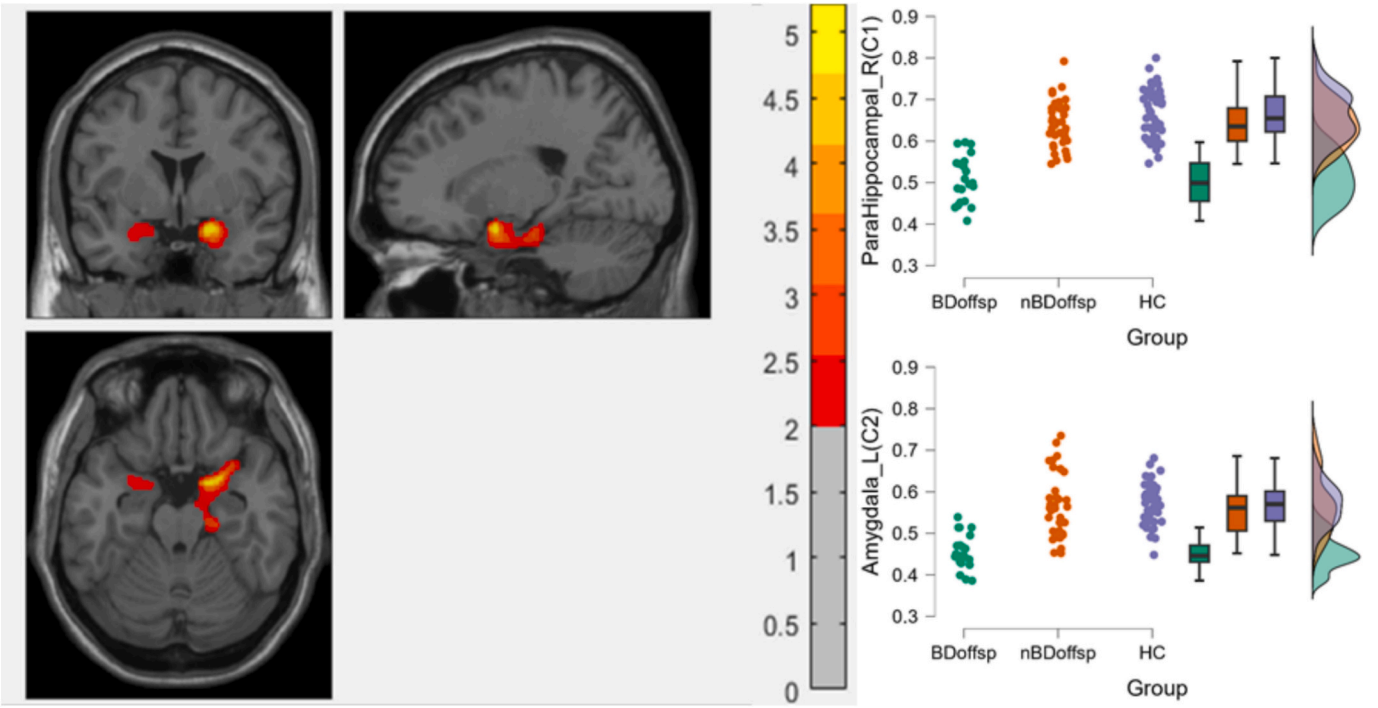


Fig. 1. The pictures display the regions that the gray matter volume difference between three groups (left). Post-hoc analysis shows that BD-offspring group have a greater lower gray matter volumes compared to both the non-BD offspring group and healthy controls (right). Note: BDoffsp = BD offspring with ADHD, nBDoffsp = non-BD offspring with ADHD, HC = healthy controls.

Table 2
Difference among groups in gray matter volume.

Anatomic region	Cluster size	Peak P	Peak MNI coordinates		
			X	Y	Z
ParaHippocampal_R & Amygdala_R	1446	<0.001	19.5	3	−18
Amygdala_L	246	0.005	−27	3	−21

Table 3
Difference among groups in cortical thickness.

Anatomic region (main/overlap region)	Cluster size	Peak P	Peak MNI coordinates		
			X	Y	Z
Precentral_R	281	<0.001	31	−22	67
Fusiform_R	105	<0.001	27	−46	−18
Parahippocampal_R	95	0.004	20	−44	−11
Temporalpole_R	70	0.006	29	22	−22
Precentral_L	210	<0.001	−31	−23	65
Fusiform_L	155	<0.001	−27	−46	−19
Temporalpole_L	151	0.001	−34	27	−19
Parahippocampal_L	150	0.004	−17	−34	−12

Our study found that BD offspring with ADHD, compared to non-BD offspring and healthy controls (HCs), exhibited decreased gray matter (GM) volume primarily in the right parahippocampal gyrus, left amygdala. The hippocampus, parahippocampal gyrus, and amygdala are all components of the limbic system, a network of interconnected cortical and subcortical brain structures that plays a central role in integrating visceral states, emotions, cognition, and behavior (Rolls, 2019). This system is involved in key functions such as memory formation and retrieval, emotional regulation, and spatial orientation. Additionally, it contributes significantly to social interactions, motivation, and mood regulation (Catani et al., 2013). The parahippocampal gyrus and hippocampus have been identified as critical regions in the interaction between emotion and memory. These areas not only support the encoding of memories related to emotional events but also facilitate the development and contextual learning of fear responses (Lang and Bradley, 2010; Phillips et al., 2008; Rich et al., 2008). The amygdala plays a pivotal role in processing emotions closely tied to survival, forming the core foundation of human motivation and emotional reactivity (LeDoux, 2000).

Interestingly, the brain regions showing gray matter (GM) volume reductions in our study bear a strong resemblance to those reported in studies of individuals at ultra-high risk for bipolar disorder, who also exhibit GM loss in the limbic system and adjacent areas (Lin et al., 2015). Collectively, these regions are strongly associated with emotional regulation and processing (Rolls, 2015), suggesting that they may represent neuroimaging markers of genetic vulnerability to bipolar disorder. In contrast, this study did not detect significant differences in GM volume between the non-BD offspring group and healthy controls (HCs). This lack of findings may be related to the relatively young age of the participants. Research suggests that ADHD is characterized by delayed brain maturation (Shaw et al., 2007), and structural differences between individuals with ADHD and typically developing controls tend to diminish with age (Hoogman et al., 2017). Moreover, several meta-analyses have failed to identify consistent structural abnormalities in the brains of individuals with ADHD, leading to the hypothesis that ADHD may reflect a dysconnectivity disorder rather than localized structural deficits in specific brain regions (Samea et al., 2019).

In the cortical thickness analysis, our study found reduced cortical thickness in the bilateral parahippocampal gyri — a finding consistent with our VBM results. The SBM analysis further revealed cortical thickness differences between the BD offspring group and both the non-BD offspring group and healthy controls (HCs) in several brain regions not identified by the VBM analysis. These included significant cortical thinning in the bilateral precentral gyri, bilateral fusiform gyri, bilateral temporopolar. The precentral gyrus, also known as the primary motor cortex, plays a critical role in motor execution and sensorimotor

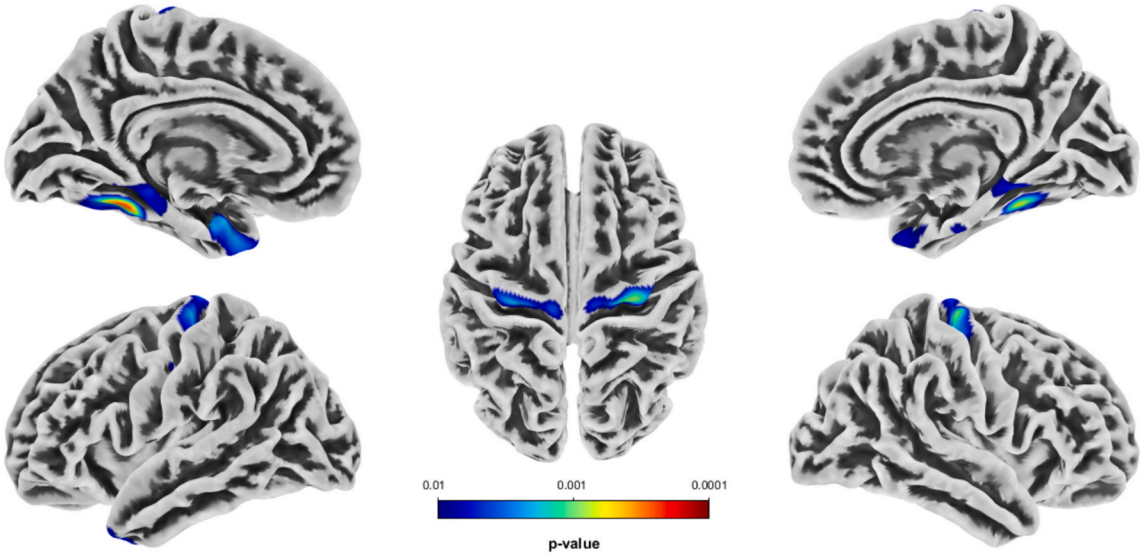


Fig. 2. The pictures display the regions that the cortical thickness difference between three groups.

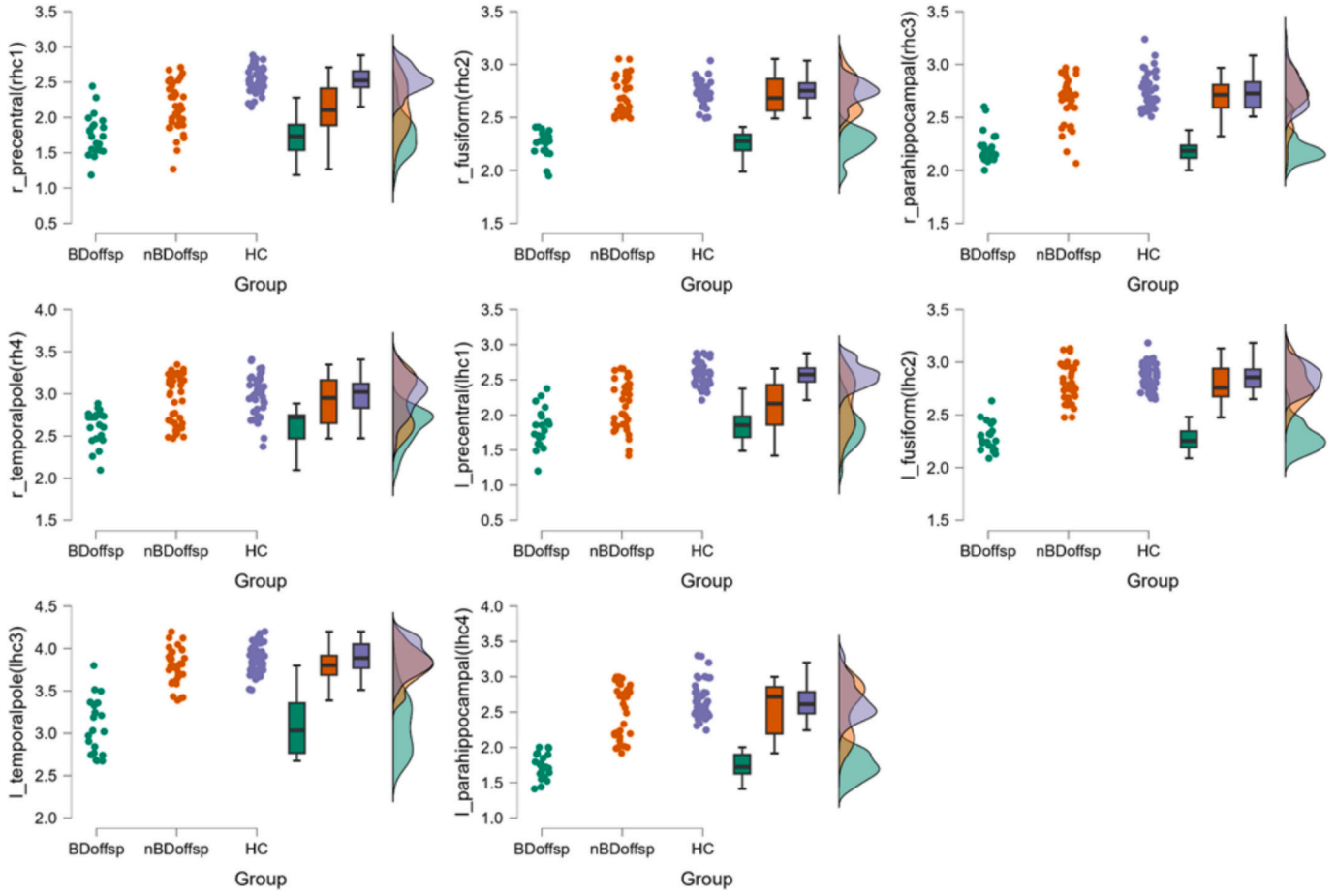


Fig. 3. The post-hoc analysis shows that BD offspring have significantly reduced cortical thickness in the above regions compare to non-BD offspring and healthy controls. Non-BD offspring have significantly reduced cortical thickness in bilateral precentral gyri compare to healthy controls. Note: BDoffsp = BD offspring with ADHD, nBDoffsp = non-BD offspring with ADHD, HC = healthy controls.

integration (Keysers et al., 2010). Dysfunction in this region has been associated with impaired motor inhibition during psychiatric states such as mania and agitation (Cummings, 1993). Reduced cortical thickness in the precentral gyrus has been reported in both individuals with BD and those with ADHD, potentially reflecting shared neurobiological impairments related to motor control and impulse regulation (İnal et al., 2023; You et al., 2024). Our findings further indicate that ADHD patients with a genetic high risk for BD exhibited higher scores on the

hyperactivity/impulsivity subscale of the SNAP-IV, which may reflect poorer motor control and more pronounced cortical thinning in the precentral gyrus. The fusiform gyrus, part of the inferior temporal cortex, is critically involved in visual cognition and facial recognition (Axmacher et al., 2008; Pierce et al., 2001). Cortical thinning in this region may contribute to altered social and emotional processing in high-risk populations. In addition, the temporopolar region integrates exteroceptive sensory information with interoceptive states, playing a crucial role in social cognition by linking external stimuli with internal affective states (Mesulam, 2023).

Recently, there has also been studies on ADHD patients with a family history of bipolar I disorder (Qin et al., 2024; Zhu et al., 2023). Interestingly, they found that patients of ADHD with a family history of bipolar disorder showed lower gray matter volumes in the limbic system, as well as cortical thickness thinning in the parahippocampal cortex, temporal pole cortex, and fusiform cortex, compared to typical ADHD patients. However, in their study, these patients additionally exhibited cortical thinning in the superior frontal, temporal, orbitofrontal, and cingulate cortices, as well as in striatal regions strongly implicated in emotional regulation which were not observed in our study. We speculate that the discrepancies between the two studies may be related to the fact that the BDO group in our study had not previously experienced mood symptoms. Alternatively, it is possible that our relatively small sample size limited our ability to detect such differences. Notably, our findings are consistent with those studies that also reported abnormal cortical thickness in the parahippocampal gyrus, fusiform gyrus, temporal regions, among individuals with bipolar disorder. Our finding further supports our perspective that offspring of individuals with BD who have ADHD differ from typical ADHD patients, and these neurostructural differences may serve as potential biomarkers of elevated risk for BD.

Finally, we also found that cortical thickness in the precentral gyrus was negatively correlated with hyperactivity/impulsivity scores in ADHD. Studies have indicated that cortical thinning in the precentral gyrus of individuals with ADHD leads to reduced GABA levels in this region, contributing to ADHD-related symptoms (Edden et al., 2012). Treatment with stimulant medications may alleviate this cortical thinning, suggesting that the persistent neurochemical and neurophysiological alterations in ADHD may be associated with this abnormality (Shaw et al., 2009). However, why offspring of individuals with both ADHD and bipolar disorder exhibit more severe cortical thinning in the precentral gyrus remains to be further investigated.

Our study has several limitations that should be considered. First, the sample size was relatively small, which may limit the ability to detect subtle differences and suggests the need for replication in larger cohorts. However, the high consistency with prior research findings supports the reliability of our results. Second, as this is a cross-sectional study, longitudinal studies are necessary to further investigate the relationship between our findings and the risk of BD progression over time. Longitudinal designs could provide more insight into the developmental trajectories and potential predictive markers of BD onset. Third, our study did not include a matched group of patients diagnosed with bipolar disorder. The inclusion of such a control group would have enhanced the validity of our data analysis and improved the clinical interpretability of our conclusions. Future research incorporating individuals with BD could offer valuable comparisons and deeper insights into the neurobiological underpinnings of genetic risk for BD among ADHD patients.

5. Conclusion

Our findings indicate that offspring of individuals with bipolar disorder who have ADHD but who have not yet exhibited mood symptoms or even subthreshold mood symptoms show brain structural alterations similar to those observed in BD, and more pronounced than those typically associated with ADHD alone. These abnormalities are primarily characterized by reduced gray matter volume in the limbic system, as well as cortical thinning in the precentral gyrus, fusiform gyrus, temporal pole, and parahippocampal gyrus. The severity of ADHD hyperactivity/impulsivity symptoms is negatively correlated with cortical thickness in the precentral gyrus. This study suggests that ADHD in the offspring of individuals with BD may differ from typical ADHD. The identified neuroanatomical changes may serve as potential biomarkers with significant clinical implications. These findings identify promising candidate biomarkers that may facilitate early identification and targeted intervention in individuals at genetic risk for BD.

CRediT authorship contribution statement

Rui Lu: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Yu Deng:** Writing – original draft, Project administration, Methodology, Data curation. **Yuchun Qin:** Writing – review & editing, Data curation. **Yuanju Lou:** Writing – review & editing. **Hehua Zeng:** Writing – review & editing. **Songtao Tan:** Data curation. **Lvqin Xu:** Data curation. **Guiyun Xu:** Writing – review & editing, Funding acquisition. **Yuanyuan Lv:** Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

We have no conflicts of interest to declare relevant to the content of this manuscript. There are no financial, personal, or professional interests that could inappropriately influence the work or its conclusions.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2025.120322.

References

Albajara Sáenz, A., Villemonteix, T., Massat, I., 2019. Structural and functional neuroimaging in attention-deficit/hyperactivity disorder. Dev. Med. Child Neurol. 61 (4), 399–405. https://doi.org/10.1111/dmcn.14050.

Axelson, D.A., Birmaher, B., Strober, M.A., Goldstein, B.I., Ha, W., Gill, M.K., Goldstein, T.R., Yen, S., Hower, H., Hunt, J.I., Liao, F., Iyengar, S., Dickstein, D., Kim, E., Ryan, N.D., Frankel, E., Keller, M.B., 2011. Course of subthreshold bipolar disorder in youth: diagnostic progression from bipolar disorder not otherwise specified. J. Am. Acad. Child Adolesc. Psychiatry 50 (10), 1001–1016.e3. https://doi.org/10.1016/j.jaac.2011.07.005.

Axmacher, N., Schmitz, D.P., Wagner, T., Elger, C.E., Fell, J., 2008. Interactions between medial temporal lobe, prefrontal cortex, and inferior temporal regions during visual working memory: a combined intracranial EEG and functional magnetic resonance imaging study. J. Neurosci. 28 (29), 7304–7312. https://doi.org/10.1523/JNEUROSCI.1778-08.2008.

Bienvenu, O.J., Davydow, D.S., Kendler, K.S., 2011. Psychiatric "diseases" versus behavioral disorders and degree of genetic influence. Psychol. Med. 41 (1), 33–40. https://doi.org/10.1017/S003329171000084X.

Catani, M., Dell'acqua, F., Thiebaut de Schotten, M., 2013. A revised limbic system model for memory, emotion and behaviour. Neurosci. Biobehav. Rev. 37 (8), 1724–1737. https://doi.org/10.1016/j.neubiorev.2013.07.001.

Cummings, J.L., 1993. Frontal-subcortical circuits and human behavior. Arch. Neurol. 50 (8), 873–880. https://doi.org/10.1001/archneur.1993.00540080076020.

Doucette, S., Levy, A., Flowerdew, G., Horrocks, J., Grof, P., Ellenbogen, M., Duffy, A., 2016. Early parent-child relationships and risk of mood disorder in a Canadian sample of offspring of a parent with bipolar disorder: findings from a 16-year prospective cohort study. Early Interv. Psychiatry 10 (5), 381–389. https://doi.org/10.1111/eip.12195.

Duffy, A., Horrocks, J., Doucette, S., Keown-Stoneman, C., McCloskey, S., Grof, P., 2014. The developmental trajectory of bipolar disorder. Br. J. Psychiatry J. Ment. Sci. 204 (2), 122–128. https://doi.org/10.1192/bjp.bp.113.126706.

Edden, R.A.E., Crocetti, D., Zhu, H., Gilbert, D.L., Mostofsky, S.H., 2012. Reduced GABA concentration in attention-deficit/hyperactivity disorder. Arch. Gen. Psychiatry 69 (7), 750–753. https://doi.org/10.1001/archgenpsychiatry.2011.2280.

Gaser, C., Dahnke, R., Thompson, P.M., Kurth, F., Luders, E., Initiative, T.A.D.N., 2024. CAT: a computational anatomy toolbox for the analysis of structural MRI data. Gigascience 13. https://doi.org/10.1093/gigascience/giae049.

GBD 2019 Adolescent Mortality Collaborators, 2021. Global, regional, and national mortality among young people aged 10-24 years, 1950-2019: a systematic analysis for the Global Burden of Disease Study 2019. Lancet 398 (10311), 1593–1618. https://doi.org/10.1016/S0140-6736(21)01546-4.

Goff, D.C., Falkai, P., Fleischhacker, W.W., Girgis, R.R., Kahn, R.M., Uchida, H., Zhao, J., Lieberman, J.A., 2017. The long-term effects of antipsychotic medication on clinical course in schizophrenia. Am. J. Psychiatry 174 (9), 840–849. https://doi.org/10.1176/appi.ajp.2017.16091016.

Hoogman, M., Bralten, J., Hibar, D.P., Mennes, M., Zwiers, M.P., Schweren, L.S.J., van Hulzen, K.J.E., Medland, S.E., Shumskaya, E., Jahanshad, N., de Zeeuw, P., Szekely, E., Sudre, G., Wolfers, T., Onnink, A.M.H., Dammers, J.T., Mostert, J.C., Vives-Gilabert, Y., Kohls, G., Franke, B., 2017. Subcortical brain volume differences in participants with attention deficit hyperactivity disorder in children and adults: a cross-sectional mega-analysis. Lancet Psychiatry 4 (4), 310–319. https://doi.org/10.1016/S2215-0366(17)30049-4.

Howes, O.D., Falkenberg, I., 2011. Early detection and intervention in bipolar affective disorder: targeting the development of the disorder. Curr. Psychiatry Rep. 13 (6), 493–499. https://doi.org/10.1007/s11920-011-0229-8.

İnal, N., Cavusoglu, B., Ermiş, Ç., Turan, S., Gormez, V., Karabay, N., 2023. Reduced cortical thicknesses of adolescents with bipolar disorder and relationship with brain-derived neurotrophic factor. Scand. J. Child Adolesc. Psychiatr. Psychol. 11 (1), 78–86. https://doi.org/10.2478/sjcapp-2023-0008.

Keysers, C., Kaas, J.H., Gazzola, V., 2010. Somatosensation in social perception. Nat. Rev. Neurosci. 11 (6), 417–428. https://doi.org/10.1038/nrn2833.

Lai, C.-H., 2021. Fronto-limbic neuroimaging biomarkers for diagnosis and prediction of treatment responses in major depressive disorder. Prog. Neuro-Psychopharmacol. Biol. Psychiatry 107, 110234. https://doi.org/10.1016/j.pnpbp.2020.110234.

Lang, P.J., Bradley, M.M., 2010. Emotion and the motivational brain. Biol. Psychol. 84 (3), 437–450. https://doi.org/10.1016/j.biopsycho.2009.10.007.

LeDoux, J.E., 2000. Emotion circuits in the brain. Annu. Rev. Neurosci. 23, 155–184. https://doi.org/10.1146/annurev.neuro.23.1.155.

Lin, K., Xu, G., Wong, N.M.L., Wu, H., Li, T., Lu, W., Chen, K., Chen, X., Lai, B., Zhong, L., So, K.-F., Lee, T.M.C., 2015. A multi-dimensional and integrative approach to examining the high-risk and ultra-high-risk stages of bipolar disorder. EBioMedicine 2 (8), 919–928. https://doi.org/10.1016/j.ebiom.2015.06.027.

Long, X., Li, L., Wang, X., Cao, Y., Wu, B., Roberts, N., Gong, Q., Kemp, G.J., Jia, Z., 2023. Gray matter alterations in adolescent major depressive disorder and adolescent bipolar disorder. J. Affect. Disord. 325, 550–563. https://doi.org/10.1016/j.jad.2023.01.049.

Lublóy, Á., Keresztúri, J.L., Németh, A., Mihalicza, P., 2020. Exploring factors of diagnostic delay for patients with bipolar disorder: a population-based cohort study. BMC Psychiatry 20 (1), 75. https://doi.org/10.1186/s12888-020-2483-y.

Merikangas, K.R., Jin, R., He, J.-P., Kessler, R.C., Lee, S., Sampson, N.A., Viana, M.C., Andrade, L.H., Hu, C., Karam, E.G., Ladea, M., Medina-Mora, M.E., Ono, Y., Posada-Villa, J., Sagar, R., Wells, J.E., Zarkov, Z., 2011. Prevalence and correlates of bipolar spectrum disorder in the world mental health survey initiative. Arch. Gen. Psychiatry 68 (3), 241–251. https://doi.org/10.1001/archgenpsychiatry.2011.12.

Mesulam, M.M., 2023. Temporopolar regions of the human brain. Brain 146 (1), 20–41. https://doi.org/10.1093/brain/awac339.

Mullins, N., Forstner, A.J., O'Connell, K.S., Coombes, B., Coleman, J.R.I., Qiao, Z., Als, T. D., Bigdeli, T.B., Børte, S., Bryois, J., Charney, A.W., Drange, O.K., Gandal, M.J., Hagenaars, S.P., Ikeda, M., Kamitaki, N., Kim, M., Krebs, K., Panagiotaropoulou, G., Andreassen, O.A., 2021. Genome-wide association study of more than 40,000 bipolar disorder cases provides new insights into the underlying biology. Nat. Genet. 53 (6), 817–829. https://doi.org/10.1038/s41588-021-00857-4.

Phillips, M.L., Ladouceur, C.D., Drevets, W.C., 2008. A neural model of voluntary and automatic emotion regulation: implications for understanding the pathophysiology and neurodevelopment of bipolar disorder. Mol. Psychiatry 13 (9), 833–857. https://doi.org/10.1038/mp.2008.65.

Pierce, K., Müller, R.A., Ambrose, J., Allen, G., Courchesne, E., 2001. Face processing occurs outside the fusiform "face area" in autism: evidence from functional MRI. Brain J. Neurol. 124 (Pt 10), 2059–2073. https://doi.org/10.1093/brain/124.10.2059.

Qin, K., Lei, D., Zhu, Z., Li, W., Tallman, M.J., Rodrigo Patino, L., Fleck, D.E., Aghera, V., Gong, Q., Sweeney, J.A., McNamara, R.K., 2024. Different brain functional network abnormalities between attention-deficit/hyperactivity disorder youth with and without familial risk for bipolar disorder. Eur. Child Adolesc. Psychiatry 33 (5). https://doi.org/10.1007/s00787-023-02245-1.

Raouna, A., Osam, C.S., MacBeth, A., 2018. Clinical staging model in offspring of parents with bipolar disorder: a systematic review. Bipolar Disord. 20 (4), 313–333. https://doi.org/10.1111/bdi.12604.

Rich, B.A., Fromm, S.J., Berghorst, L.H., Dickstein, D.P., Brotman, M.A., Pine, D.S., Leibenluft, E., 2008. Neural connectivity in children with bipolar disorder: impairment in the face emotion processing circuit. J. Child Psychol. Psychiatry 49 (1), 88–96. https://doi.org/10.1111/j.1469-7610.2007.01819.x.

Rolls, E.T., 2015. Limbic systems for emotion and for memory, but no single limbic system. Cortex 62, 119–157. https://doi.org/10.1016/j.cortex.2013.12.005.

Rolls, E.T., 2019. The cingulate cortex and limbic systems for emotion, action, and memory. Brain Struct. Funct. 224 (9), 3001–3018. https://doi.org/10.1007/s00429-019-01945-2.

Samea, F., Soluki, S., Nejati, V., Zarei, M., Cortese, S., Eickhoff, S.B., Tahmasian, M., Eickhoff, C.R., 2019. Brain alterations in children/adolescents with ADHD revisited: a neuroimaging meta-analysis of 96 structural and functional studies. Neurosci. Biobehav. Rev. 100, 1–8. https://doi.org/10.1016/j.neubiorev.2019.02.011.

Selvaraj, S., Arnone, D., Job, D., Stanfield, A., Farrow, T.F., Nugent, A.C., Scherk, H., Gruber, O., Chen, X., Sachdev, P.S., Dickstein, D.P., Malhi, G.S., Ha, T.H., Ha, K., Phillips, M.L., McIntosh, A.M., 2012. Grey matter differences in bipolar disorder: a meta-analysis of voxel-based morphometry studies. Bipolar Disord. 14 (2), 135–145. https://doi.org/10.1111/j.1399-5618.2012.01000.x.

Shaw, P., Eckstrand, K., Sharp, W., Blumenthal, J., Lerch, J.P., Greenstein, D., Clasen, L., Evans, A., Giedd, J., Rapoport, J.L., 2007. Attention-deficit/hyperactivity disorder is characterized by a delay in cortical maturation. Proc. Natl. Acad. Sci. USA 104 (49), 19649–19654. https://doi.org/10.1073/pnas.0707741104.

Shaw, P., Sharp, W.S., Morrison, M., Eckstrand, K., Greenstein, D.K., Clasen, L.S., Evans, A.C., Rapoport, J.L., 2009. Psychostimulant treatment and the developing cortex in attention deficit hyperactivity disorder. Am. J. Psychiatry 166 (1), 58–63. https://doi.org/10.1176/appi.ajp.2008.08050781.

Song, J., Bergen, S.E., Kuja-Halkola, R., Larsson, H., Landén, M., Lichtenstein, P., 2015. Bipolar disorder and its relation to major psychiatric disorders: a family-based study in the Swedish population. Bipolar Disord. 17 (2), 184–193. https://doi.org/10.1111/bdi.12242.

Türközer, H.B., Guvenek-Cokol, P.E., Millman, Z.B., Dickstein, D., Öngür, D., 2025. Familial high-risk for psychosis and bipolar disorder amongst youth in an academic healthcare system: a real-world assessment of clinical characteristics and implications for clinical care. Early Interv. Psychiatry 19 (2), e70019. https://doi.org/10.1111/eip.70019.

Vieta, E., Salagre, E., Grande, I., Carvalho, A.F., Fernandes, B.S., Berk, M., Birmaher, B., Tohen, M., Suppes, T., 2018. Early intervention in bipolar disorder. Am. J. Psychiatry 175 (5), 411–426. https://doi.org/10.1176/appi.ajp.2017.17090972.

Vigo, D., Thornicroft, G., Atun, R., 2016. Estimating the true global burden of mental illness. Lancet Psychiatry 3 (2), 171–178. https://doi.org/10.1016/S2215-0366(15)00505-2.

Wiltink, S., Velthorst, E., Nelson, B., McGorry, P.M., Yung, A.R., 2015. Declining transition rates to psychosis: the contribution of potential changes in referral pathways to an ultra-high-risk service. Early Interv. Psychiatry 9 (3), 200–206. https://doi.org/10.1111/eip.12105.

You, W., Li, Q., Chen, L., He, N., Li, Y., Long, F., Wang, Y., Chen, Y., McNamara, R.K., Sweeney, J.A., DelBello, M.P., Gong, Q., Li, F., 2024. Common and distinct cortical thickness alterations in youth with autism spectrum disorder and attention-deficit/hyperactivity disorder. BMC Med. 22 (1), 92. https://doi.org/10.1186/s12916-024-03313-2.

Zhu, Z., Zhao, Y., Wen, K., Li, Q., Pan, N., Fu, S., Li, F., Radua, J., Vieta, E., Kemp, G.J., Biswa, B.B., Gong, Q., 2022. Cortical thickness abnormalities in patients with bipolar disorder: a systematic review and meta-analysis. J. Affect. Disord. 300, 209–218. https://doi.org/10.1016/j.jad.2021.12.080.

Zhu, Z., Lei, D., Qin, K., Tallman, M.J., Patino, L.R., Fleck, D.E., Gong, Q., Sweeney, J.A., DelBello, M.P., McNamara, R.K., 2023. Cortical and subcortical structural differences in psychostimulant-free ADHD youth with and without a family history of bipolar I disorder: a cross-sectional morphometric comparison. Transl. Psychiatry 13 (1), 368. https://doi.org/10.1038/s41398-023-02667-0.